PV101 Day 7 Daily Run Sheet

Instructor-facing plan for Rich or a substitute | Wednesday, July 15, 2026

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Topic: Load Assessment and System Sizing

Class setup: 6 students | 10:00 AM-3:00 PM | Calm, practical, evidence-first instruction

Teaching frame

- Main message: loads drive system sizing, and units make the reasoning inspectable.
- Day 7 is guided practice, not a project, proposal, capstone, or presentation.
- Use only the documented training values on the worksheet; do not invent equipment or product specifications.
- Safety remains part of design: protection, disconnects, labels, and verification points must appear on the one-line diagram.

Substitute quick start

- [ ] Read the full schedule before students arrive.
- [ ] Locate emergency disconnects, first-aid supplies, calculator policy, and secure test storage.
- [ ] Print student materials and keep answer keys/instructor notes separate.
- [ ] Write the day goal, schedule, and stop-work rule on the board.
- [ ] If uncertain about equipment or assessment handling, use the reduced-scope path and contact program leadership.

Learning targets

- Calculate daily energy use in watt-hours from a documented loads list.
- Use provided assumptions to calculate basic battery Wh/Ah and array watts.
- Choose an inverter rating from documented simultaneous AC loads and headroom.
- Draw and label a one-line system path with protection, disconnects, and verification points.
- Check units, assumptions, and reasonableness before instructor signoff.

Materials

- Chapters 11-12 quiz and key
- Day 7 slide deck and whiteboard outline
- Student worksheet and guided-practice guide
- Calculators, pencils, rulers, scrap paper
- Documented RV training loads and provided assumptions
- Evidence checklist and reflection sheet

Daily schedule

Time	Block	Instructor notes
10:00-10:10	Arrival, attendance, check-in	Students get notebooks and Chapters 11-12 quiz. Instructor displays: Measure. Verify. Document. Then proceed.
10:10-10:20	Safety reset + day goal	Review stop-work authority and explain that today is guided sizing practice, not a project or product proposal.
10:20-10:35	Morning quiz	Individual Chapters 11-12 reading check. Ten questions; quiet work.
10:35-10:55	Quiz correction	Correct together in a different color. Review current-source verification, AHJ/permitting roles, units, and safest-answer thinking.
10:55-11:20	Whiteboard review	Connect Chapters 1-12: loads drive sizing; safety and verification constrain every decision.
11:20-11:30	Break	Reset attention. Calculators, worksheets, and pencils staged after break.
11:30-12:00	Slides + worked example	Model watts x hours, total Wh/day, battery Wh/Ah, array watts, and inverter headroom using the provided training values.
12:00-12:30	Guided example check	Students complete one example with instructor pauses after each calculation. Check units before accepting answers.
12:30-1:00	Lunch	Worksheets remain on tables. No kit wiring or energized work.
1:00-1:15	Practice introduction	Explain the RV training load worksheet, required assumptions, evidence boxes, and instructor checkpoints.

1:15-1:50	Guided worksheet: loads + battery	Students calculate daily Wh and basic storage. Instructor checks totals before array sizing.
1:50-2:15	Guided worksheet: array + inverter	Students calculate array watts and identify concurrent AC load plus headroom.
2:15-2:30	Diagram + protection points	Students draw a one-line system path and mark conceptual protection, disconnects, labels, and meter points.
2:30-2:45	Verification + revision	Peer checks math, units, assumptions, and diagram. Instructor signs only after corrections.
2:45-2:55	Reflection + evidence upload	Complete reflection and upload/submit the worksheet as directed.
2:55-3:00	Cleanup + dismissal	Calculators returned, papers collected, tables clear, next-day final reminder given.

Safety and instructional stop conditions

- A student uses an assumption or specification not provided in the worksheet.
- Units are missing or mixed without conversion.
- A sizing answer is copied without work or cannot be explained.
- The diagram omits protection, disconnects, or return paths.
- Discussion drifts into physical wiring; this day is paper/calculator practice.

Reduced-scope backup

Day 7: if calculators or printing fail, complete one class example on the whiteboard and have students show units and reasonableness on notebook paper. Day 8: if final-test printing fails, run structured review and reflection; do not improvise a different final from memory. Reschedule the secured assessment.

Early finishers

Day 7: check every unit, redo one calculation independently, and write one sentence explaining each sizing result. Day 8: quietly review answers, then begin the reflection only after turning in the final.

Instructor notes after class

What landed:

_____

_____

What confused students:

_____

_____

Equipment/material issues:

_____

_____

Carry-forward:

_____

_____